

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

Agriculture is one of the most important sectors supporting the economy and food production worldwide. Efficient water management is essential for improving crop productivity and ensuring sustainable agricultural development. Traditional irrigation methods often depend on manual observation and fixed watering schedules, which can result in over-irrigation, under-irrigation, and significant water wastage.

With the advancement of Internet of Things (IoT) technologies, modern agriculture is moving toward intelligent monitoring and automation systems. Smart irrigation systems use sensors, microcontrollers, communication networks, and cloud technologies to monitor environmental conditions and automatically control irrigation activities. These systems help optimize water consumption, reduce labor requirements, and improve crop health.

This chapter reviews existing research and technologies related to smart irrigation and plant monitoring systems.

2.2 Evolution of Irrigation Systems

Irrigation systems have evolved significantly over time.

Traditional Irrigation

Traditional irrigation methods include:

- Flood Irrigation
- Furrow Irrigation
- Basin Irrigation
- Manual Watering

Although these methods are simple, they suffer from several disadvantages:

- High water wastage
- Uneven water distribution
- Increased labor requirements
- Lack of environmental monitoring
- Low efficiency

Modern Irrigation

Modern irrigation techniques include:

- Drip Irrigation
- Sprinkler Irrigation
- Smart Irrigation Systems

These systems focus on delivering the required amount of water directly to plant roots, thereby improving efficiency and reducing water losses.

2.3 Internet of Things (IoT) in Agriculture

The Internet of Things (IoT) refers to a network of interconnected devices capable of collecting, exchanging, and processing data through the internet.

In agriculture, IoT technology enables:

- Real-time monitoring
- Remote control of equipment
- Environmental data collection
- Automated irrigation
- Predictive analysis

IoT-based farming systems improve productivity by continuously monitoring environmental conditions and taking corrective actions automatically.

Major components of IoT agriculture systems include:

1. Sensors
2. Microcontrollers
3. Communication Modules
4. Cloud Platforms
5. User Interfaces

The proposed smart irrigation system utilizes ESP32 as the processing unit and environmental sensors for data acquisition.

2.4 Soil Moisture Monitoring Systems

Soil moisture is one of the most critical factors affecting plant growth and crop yield.

Several researchers have developed systems that continuously monitor soil moisture and control irrigation accordingly.

Soil Moisture Sensors

Soil moisture sensors measure the amount of water present in soil.

Common types include:

1. Resistive Sensors
2. Capacitive Sensors

The sensor used in this project provides analog output values which are converted into moisture percentages using calibration techniques.

Advantages of soil moisture monitoring:

- Prevents over-irrigation
 - Saves water
 - Improves plant growth
 - Reduces energy consumption
-

2.5 Temperature and Humidity Monitoring

Environmental conditions such as temperature and humidity directly influence plant growth.

Researchers commonly use sensors like:

- DHT11
- DHT22
- SHT31

These sensors provide:

- Ambient temperature
- Relative humidity

Monitoring these parameters helps farmers understand environmental conditions and make better irrigation decisions.

The DHT11 sensor was selected for this project due to:

- Low cost
 - Ease of interfacing
 - Reliable performance
 - Adequate accuracy for educational applications
-

2.6 ESP32-Based Smart Agriculture Systems

ESP32 is one of the most popular microcontrollers for IoT applications.

Features of ESP32 include:

- Dual-core processor
- Built-in Wi-Fi
- Built-in Bluetooth
- Multiple GPIO pins
- ADC support
- Low power consumption

Several research studies have demonstrated the effectiveness of ESP32 in:

- Smart irrigation
- Weather monitoring
- Greenhouse automation
- Crop monitoring

The ESP32 provides a powerful and cost-effective platform for developing intelligent agricultural solutions.

2.7 Automated Irrigation Systems

Automated irrigation systems use sensors and controllers to determine when irrigation is required.

The general working process includes:

1. Monitoring soil moisture
2. Comparing readings with threshold values
3. Activating irrigation if necessary
4. Stopping irrigation after adequate moisture is achieved

Benefits include:

- Reduced human intervention
- Water conservation
- Improved crop productivity
- Consistent irrigation scheduling

The proposed project follows the same principle by controlling a relay-operated water pump.

2.8 OLED Display-Based Monitoring Systems

Display units provide immediate feedback regarding system status.

Common displays used in IoT systems include:

- LCD Displays
- OLED Displays
- TFT Displays

OLED displays offer several advantages:

- Low power consumption
- High contrast
- Compact size
- Easy I2C communication

In this project, the OLED display presents:

- Temperature
- Humidity
- Soil Moisture Percentage
- Pump Status

This enables real-time monitoring without requiring external devices.

2.9 Relay-Controlled Water Pump Systems

Most automated irrigation systems require a mechanism to control water flow.

Relay modules are commonly used because:

- They provide electrical isolation.
- They can switch higher-current devices.
- They are easy to interface with microcontrollers.

The relay module receives signals from ESP32 and controls the water pump accordingly.

Advantages include:

- Safe operation
 - Reliable switching
 - Low cost
 - Simple implementation
-

2.10 Existing Research Work

Several researchers have proposed smart irrigation systems using Arduino, NodeMCU, Raspberry Pi, and ESP32.

Study 1

Researchers developed a soil moisture-based irrigation system using Arduino Uno.

Results:

- Reduced water consumption.
- Improved irrigation efficiency.

Limitations:

- No wireless connectivity.

Study 2

A NodeMCU-based irrigation system was developed with cloud monitoring.

Results:

- Remote monitoring capability.
- Mobile application support.

Limitations:

- Increased complexity.

Study 3

ESP32-based smart farming systems integrated multiple sensors and cloud storage.

Results:

- Real-time monitoring.
- Remote access.
- Data analytics.

Limitations:

- Higher implementation cost.

The proposed project focuses on a low-cost implementation suitable for educational and small-scale agricultural applications.

2.11 Comparative Analysis

Parameter	Traditional Irrigation	Smart Irrigation
Water Usage	High	Low
Monitoring	Manual	Automatic
Labor Requirement	High	Low
Cost Efficiency	Moderate	High
Accuracy	Low	High
Automation	No	Yes
Data Collection	No	Yes

The comparison clearly demonstrates the advantages of smart irrigation systems over traditional approaches.

2.12 Research Gap

Although many smart irrigation systems have been proposed, several challenges still exist:

- High implementation cost
- Complex cloud integration
- Dependence on internet connectivity
- Maintenance requirements

The proposed system addresses these challenges by developing a low-cost, easy-to-implement solution using ESP32 and basic sensors while maintaining automation capabilities.

2.13 Summary

This chapter reviewed existing irrigation technologies, IoT applications in agriculture, environmental monitoring systems, and automated irrigation solutions. The literature survey highlights the growing importance of smart irrigation systems in improving agricultural productivity and water conservation.

Based on the findings, ESP32 combined with soil moisture and environmental sensors provides an effective platform for developing intelligent irrigation systems. The proposed project aims to implement these concepts in a practical, low-cost, and scalable manner suitable for educational and agricultural applications.